Example 9.1

Fit a trend line by the method of freehand method for the given data.

Year	2000	2001	2002	2003	2004	2005	2006	2007
Sales	30	46	25	59	40	60	38	65

Solution:

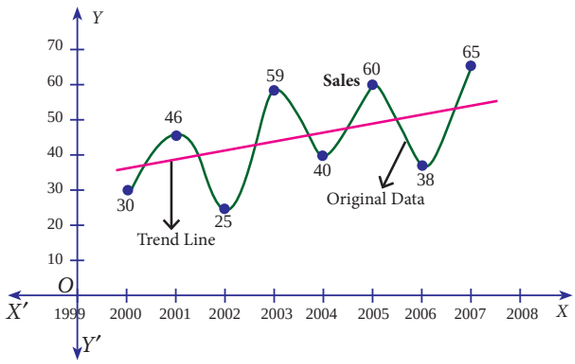


Fig: 9.1

Note

The trend drawn by the freehand method can be extended to predict the future values of the given data. However, this method is subjective in nature, predictions obtained by this method depends on the personal bias and judgement of the investigator handling the data.

(ii) Method of Semi-Averages

In this method, the semi-averages are calculated to find out the trend values. Now, we will see the working procedure of this method.

Procedure:

(i) The data is divided into two equal parts. In case of odd number of data, two equal parts can be made simply by omitting the middle year.

(ii) The average of each part is calculated, thus we get two points.

(iii) Each point is plotted at the mid-point (year) of each half.

(iv) Join the two points by a straight line.

(v) The straight line can be extended on either side.

(vi) This line is the trend line by the methods of semi-averages.

Example 9.2

Fit a trend line by the method of semi-averages for the given data.

Year	2000	2001	2002	2003	2004	2005	2006
Production	105	115	120	100	110	125	135

Solution:

Since the number of years is odd(seven), we will leave the middle year's production value and obtain the averages of first three years and last three years.

Year	Production	Average
2000	105	
2001	115	$\frac{105+115+120}{3}=113.33$
2002	120	
2003	100 (left out)	
2004	110	
2005	125	$\frac{110+125+135}{3}=123.33$
2006	135	

Table 9.1

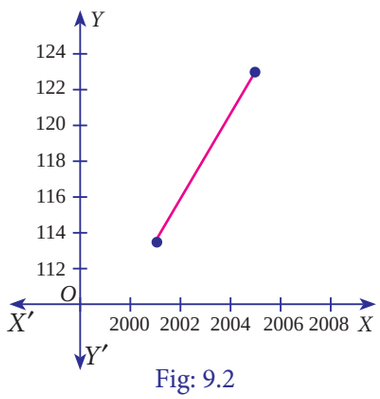


Fig: 9.2

Example 9.3

Fit a trend line by the method of semi-averages for the given data.

Year	1990	1991	1992	1993	1994	1995	1996	1997
Sales	15	11	20	10	15	25	35	30

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